The Applications of AWS Blockchain Templates

Blockchain technology is a recent financial technology that has completely transformed business transactions. The records of the ledger databases it provides are immutable and cryptographically signed using a distributed consensus or validation protocol. This has made the blockchain popular for executing transactions in multi-party business environments. It guarantees the authenticity and non-tampering of transactions without the need for any centralised authority. This article focuses on the Amazon Web Services (AWS) blockchain templates and their applications.

One of the most explosive innovations involving blockchain is the cloud based blockchain platform. For greater access, flexibility, efficiency, and security, cloud computing is evolving to a decentralised infrastructure. Innovations like serverless technology are causing a paradigm shift in computing, freeing systems administrators for designing and configuring as well as managing the servers and resources required to run business applications.

Decentralised frameworks also allow business owners to utilise only the storage they need. With centralised cloud computing storage, this isn't always the case. With the implementation of decentralisation in networks, it's easier to ensure that only authorised users have access to stored data. Security through this type of network is more extensive because it's divided between several different storage components. If there is a breach in a decentralised system, no single user can have access to all your data since it's scattered across the world when it's stored. A blockchain cloud storage solution takes a user's data and breaks it into tiny chunks of data. It then takes these chunks of data, encrypts them to add an extra layer of security, and distributes them throughout the network.

This is done by using blockchain features such as transaction ledgers, cryptographic hash functions, and public/private key encryption. The data that is being stored throughout the blockchain ledger is split up and hashed a significant number of times. More importantly, the different chunks of hashed data are not stored on the same storage device. These are split randomly throughout the network. This ensures that even if one bad actor is able to decrypt a piece of data, the entire file cannot be accessed. The node would not be able to identify who the data belongs to, and hence the true user will remain anonymous.